

Assignment No 4

Smart Grievance Hnadling System

Table Setup

```
mysql> CREATE TABLE Issue (
->     issue_id INT PRIMARY KEY AUTO_INCREMENT,
->     title VARCHAR(255) NOT NULL,
->     description TEXT,
->     area VARCHAR(100),
->     severity ENUM('Low', 'Medium', 'High') DEFAULT 'Low',
->     priority_score INT DEFAULT 0,
->     status ENUM('Open', 'In Progress', 'Resolved') DEFAULT 'Open',
->     report_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
->     resolution_date TIMESTAMP NULL,
->     citizen_id INT NOT NULL,
->     FOREIGN KEY (citizen_id) REFERENCES Citizen(citizen_id) ON DELETE CASCADE,
->     INDEX(priority_score)
[ -> );
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> CREATE TABLE Issue_Priority_Log (
->     log_id INT PRIMARY KEY AUTO_INCREMENT,
->     issue_id INT NOT NULL,
->     old_priority INT,
->     new_priority INT,
->     change_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
->     FOREIGN KEY (issue_id) REFERENCES Issue(issue_id) ON DELETE CASCADE
[ -> );
Query OK, 0 rows affected (0.02 sec)
```

1) INSERT Trigger

```
mysql> CREATE TRIGGER trg_after_issue_insert
-> AFTER INSERT ON Issue
-> FOR EACH ROW
-> BEGIN
->     INSERT INTO Issue_Priority_Log
->     (issue_id, new_priority)
->     VALUES
->     (
->         NEW.issue_id,
->         NEW.priority_score
->     );
[ -> END //
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> INSERT INTO Issue
-> (title, description, area, severity, citizen_id)
-> VALUES
-> (
->   'Sewage overflow',
->   'Drain water overflowing on road',
->   'Pimpri',
->   'High',
->   3
[   -> );
Query OK, 1 row affected (0.00 sec)
```

```
mysql> SELECT *
-> FROM Issue_Priority_Log
[   -> ORDER BY log_id DESC;
```

log_id	issue_id	old_priority	new_priority	change_date
12	16	NULL	0	2026-03-29 22:36:33
9	9	0	60	2026-03-26 18:05:20
8	8	0	90	2026-03-26 18:05:20
7	7	0	30	2026-03-26 18:05:20
6	6	0	90	2026-03-26 18:05:20
5	5	0	60	2026-03-26 18:05:20
4	4	0	30	2026-03-26 18:05:20
3	3	0	90	2026-03-26 18:05:20
2	2	0	60	2026-03-26 18:05:20
1	1	0	90	2026-03-26 18:05:20

```
10 rows in set (0.00 sec)
```

2) DELETE Trigger

```
mysql> CREATE TRIGGER trg_after_issue_delete
-> AFTER DELETE ON Issue
-> FOR EACH ROW
-> BEGIN
->   INSERT INTO Issue_Priority_Log
->   (issue_id, old_priority)
->   VALUES
->   (
->     OLD.issue_id,
->     OLD.priority_score
->   );
[   -> END //
```

```
Query OK, 0 rows affected (0.01 sec)
```

```
[mysql> DELETE FROM Issue WHERE issue_id = 12;
Query OK, 1 row affected (0.01 sec)
```

```
mysql> SELECT *
-> FROM Issue_Priority_Log
-> ORDER BY log_id DESC;
```

log_id	issue_id	old_priority	new_priority	change_date
14	12	0	NULL	2026-03-29 22:40:20
12	16	NULL	0	2026-03-29 22:36:33
9	9	0	60	2026-03-26 18:05:20
8	8	0	90	2026-03-26 18:05:20
7	7	0	30	2026-03-26 18:05:20
6	6	0	90	2026-03-26 18:05:20
5	5	0	60	2026-03-26 18:05:20
4	4	0	30	2026-03-26 18:05:20
3	3	0	90	2026-03-26 18:05:20
2	2	0	60	2026-03-26 18:05:20
1	1	0	90	2026-03-26 18:05:20

```
11 rows in set (0.00 sec)
```

3) UPDATE Trigger

```
mysql> CREATE TRIGGER trg_after_issue_update
-> AFTER UPDATE ON Issue
-> FOR EACH ROW
-> BEGIN
-> IF OLD.priority_score != NEW.priority_score THEN
-> INSERT INTO Issue_Priority_Log
-> (issue_id, old_priority, new_priority)
-> VALUES
-> (
-> OLD.issue_id,
-> OLD.priority_score,
-> NEW.priority_score
-> );
-> END IF;
-> END //
```

Query OK, 0 rows affected (0.01 sec)

```
[mysql> DELIMITER ;
mysql> UPDATE Issue
    -> SET severity = 'High'
[    -> WHERE issue_id = 4;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT *
    -> FROM Issue
[    -> WHERE issue_id = 4;
+-----+-----+-----+-----+-----+
| issue_id | title                | description          | area      | severity |
+-----+-----+-----+-----+-----+
|         4 | Street light not working | Dark area at night | Chinchwad | High     |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)
```